Hill Choy

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SUMMARY

AI engineer with an MSc from UCL and hands-on experience building and deploying machine learning systems end to end, from prototyping through to production. Strong in Python with a solid grounding in ML fundamentals and modern software engineering practices, and a fast-learning builder genuinely excited by applying AI, including modern vision-language and generative models, to real-world problems..

EDUCATION

University College London (UCL)
MSc Computer Graphics, Vision and Imaging (***Current Average: 79%***) — *Sept 2025 - Present*
- **Highlighted Coursework**: Machine Vision, ML for Visual Computing, Inverse Problems, Robot Vision.

Hong Kong University of Science and Technology (HKUST)
BEng in Computer Engineering (***Second Class Honors, Division 1***) — *Sept 2020 - Jul 2024*
- **Highlighted Coursework**: Intro to NLP, Computer & Communication Networks.

IELTS Academic
8.0 (Reading: 8.5, Listening: 9.0, Writing: 7.0, Speaking: 7.0)

RELEVANT EXPERIENCE

Computer Vision for Automated Textile Colour Sorting — **United Kingdom**
*UCL, MSc Industry Dissertation Project (Partner: **Zori Tex**)* — *June 2026 – Present*
- Engineered an end-to-end pipeline for dominant-colour classification of multi-coloured fabrics, benchmarking 8 classical and deep architectures (colour-distance baselines, CNNs, vision transformers) on a heavily imbalanced real-world dataset to identify the most deployment-ready approach for automated textile sorting.
- Conducted extensive failure-mode analysis showing errors concentrate in inherently ambiguous, multi-coloured fabrics, with ongoing work extending the pipeline to multi-label colour prediction.

Research Assistant — **Hong Kong**
Prof. Shueng-Han Gary Chan, CSE, HKUST — *Aug 2024 - Jun 2025*
- **K11 MUSEA Cars and Pedestrian Flow Monitoring System**
  - Engineered an end-to-end analytics system tracking 60,000+ daily visitors, utilizing OpenCV and a PyTorch-based YOLO model for real-time local flow counting on edge platforms.
  - Developed a multithreaded Flask backend utilizing Redis for real-time data synchronization and SQLite/Pandas for robust storage, powering a web dashboard to optimize mall operations.
- **Privacy-Preserving Elderly Monitoring with a Novel Edge AI Camera**
  - Contributed to the development of a privacy-preserving Edge AI fall-detection system deployed on an NPU-accelerated OrangePi, utilizing Human Pose Estimation for real-time patient monitoring.
  - Developed rule-based post-processing algorithms to minimize false alarms without compromising true positive cases, achieving 98% accuracy and securing the Silver Award at the Hong Kong ICT Awards 2024.
- **Smart Traffic Management System (Kwun Tong, Hong Kong)**
  - Trained and deployed YOLO models on a dual-RTX 4090 server to process live streams from 33 cameras across 11 major junctions, monitoring 20,000+ daily vehicles per junction.
  - Engineered automated client scripts using Python requests to securely transmit vehicle counts and yellow-box stop detections to a centralized API, handling Keycloak OAuth2 token authentication.

Advanced Video Analytics and Edge AI for Smart Carpark Systems — **Hong Kong**
*HKUST, Bachelor Degree Final Year Project (Supervisor: **Prof. Shueng-Han Gary Chan, CSE, HKUST**)* — *Sept 2023 - Apr 2024*
- Designed an Edge AI smart car park system to track real-time space availability, utilizing OpenCV for video frame grabbing and preprocessing to handle complex feeds from custom low-light and wide-angle cameras.
- Deployed a PyTorch-based YOLOv8 model on a Raspberry Pi by integrating a MobileNetV3 backbone, converting to ONNX format, and utilizing Ultralytics for custom dataset annotation, halving inference latency and reducing parameters by 30% while maintaining baseline accuracy.

Intern (Summer full-time, then part-time) — **Hong Kong**
Hong Kong Telecom (HKT), Integrated Service Provision Center — *June 2023 - April 2025*
- Engineered a Python GUI executable to replace a legacy system and eliminate manual checks, automating both Excel and physical report generation for phone line data.
- Bypassed the lack of a native API using a custom web-scraping pipeline, dynamically extracting interface records to synchronize data with a MySQL database.